

AXIOMALITY

We inspect robot training data before labs pay for it, and sign the result.

The inspection record a lab accepts at intake: a verdict per episode, a reason for every reject, and a signed passport anyone can verify offline. Prototype runs on public data; no customers yet.

Working prototype · see slide 5

AXIOMALITY DATA PASSPORT

✓ SIGNATURE VALID

AX-DEMO-109 - SYNTHETIC pick-and-place episodes derived from sample 109 (demo)

CONTENT HASH	sha256:d9bc1e619e2ed73b679479011229ab13...
ACCEPTED SPLIT	147 episodes, 24,654 frames
VERDICTS	submitted 200 PASS 137 REPAIR 10 REJECT 53 UNKNOWN 0
RULEBOOK	ax-kernel-0.1.0-demo (sha256:b3c3fed10488fd8f...)
REAL / SYNTHETIC	0 real / 147 synthetic (100% synthetic demo data)
SEED SAMPLE	pointclouds/109.ply (sha256:e52300258a4e...)
ENGINE / CHECKS	ax-engine-0.1.0-prototype z3 5.1.0 13 rule families
ISSUED / KEY	2026-09-26T06:36:10+00:00 ed25519:581e68418d4a4255 (demo key)

Verify offline: python verify_passport.py (Ed25519 signature + recomputed content hash)

Synthetic demo | not a safety certification

Real prototype output: a signed passport over synthetic episodes seeded from the company's measured scan geometry

41 real episodes

checked across DROID and two other public datasets; found a stuck arm, an over-limit joint, a mis-documented field

Signed

Change any covered value and verification fails (demo key today)

Offline

Buyers verify with a public key, no call to us

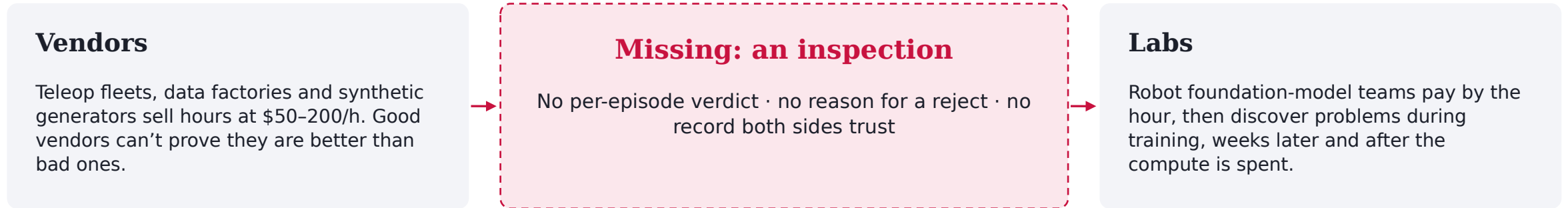
In your VPC

The engine runs where the data lives; we never hold it

PROBLEM

Labs buy robot data blind; vendors can't prove what they sold

Acceptance today is a sample video and a spot check



89%

of visual-AI teams blame data for model failures (Voxel51, 2026)

58,000+

datasets on the LeRobot hub; none validated or signed

\$50–200 / h

teleop data, sold with no record of what is usable

48%

of visual-AI teams cite data-quality problems (Voxel51, 2026)

What goes wrong is not exotic: mislabeled objects, phases out of order, a gripper that “grasps” while open, objects that teleport or float, clocks that run backwards. Rule scripts catch the easy half. The other half needs to know what the scene means.

In 2026 robot data became a traded commodity, with no inspector

Vendors are scaling, generators are flooding training, and buyers are writing big checks



Vendors scaled

- Scale: physical-AI engine past 150k hours; neutrality questioned after Meta
- Mecka: ~\$500M round in talks (Sep 2026)
- XDOF: ~\$1.2B valuation (Sep 2026)
- Build AI: ~1M hours of factory video released free (Apr 2026)



Generators arrived

- World Labs' Atlas (Sep 2026) turns phone video into robot-camera data
- NVIDIA's data-factory blueprint scores generated clips for free
- More synthetic data means more data entering training that no one inspected



Buyers are spending

- \$18.6B physical-AI venture funding in Q2 2026 alone
- Physical Intelligence (~\$11B), Skild (\$14B), Generalist (\$3B), Figure (\$39B) spend heavily on data
- Our hypothesis: buyers want a neutral referee at intake

Regulation is a later tailwind, not the reason: EU AI Act obligations for AI embedded in machinery apply from 2 Aug 2028 (Reg. 2026/1744).

Episodes in; a verdict per episode and a signed passport out

Three check families. On today's public data only the first runs fully; next, we derive object state from video so the other two run on real data.



Is the record intact?

Monotonic clocks, dropped frames, NaNs, joint limits, required metadata

Scripts do this; so do we



Is the motion possible?

Interpenetration, unsupported objects, teleports, gripper-to-object distance

Needs scene geometry and relations



Does the story contradict itself?

Task-phase order, gripper state vs contact events, object class vs affordance

Needs a rulebook proven contradiction-free

Valid failures are kept

A cup that really slips is valuable failure data, not bad data. We never rewrite a real failure into a success.

Signed, offline-verifiable

The passport binds the data hash and rulebook version. Seller, buyer or anyone else verifies with a public key.

Runs in your environment

Data never leaves the customer's VPC. We ship the engine and the rulebook; we never own the data.

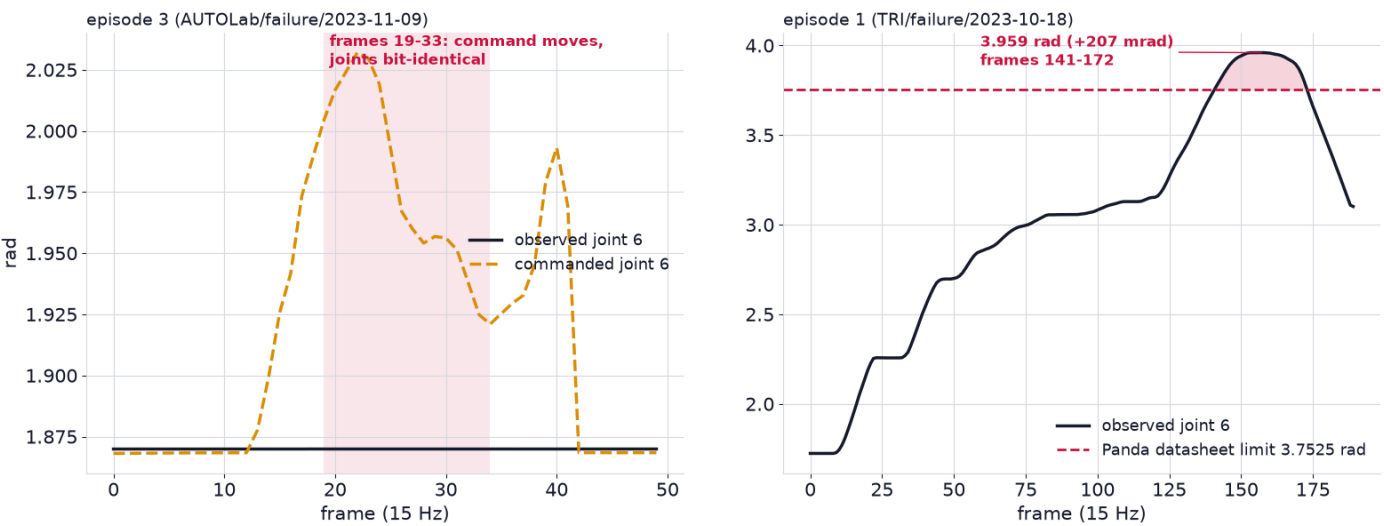
Verdict	Meaning	What happens
Accept	All three checks pass	Billable and trainable
Repair	Correctable recording defect (e.g., timestamps out of order)	New derived version; original kept
Reject	Violates a rule; frame and rule ID attached	Back to the vendor as a re-collect list
Unknown	Inputs needed for the check are missing	Says exactly what is missing; never counted as a pass

On real DROID data: a stuck arm, a mis-documented field, a possible arm mismatch

Could a script find these? Yes. We add doing it on every batch, signing the result, and listing what we could not check. 41 episodes from DROID, Jaco Play, NYU ROT.

Two real findings in DROID (droid_100, Franka Panda)

REAL data. Left: commanded motion not executed (REJECT, action-state family).
Right: joint 6 beyond the Panda datasheet limit (REJECT; the arm could be an FR3, whose joint-6 limit is 4.52 rad).



Stuck arm / frozen stream

DROID ep. 3, frames 19–33: the commanded joint position moves (median 0.133 rad) while the measured joints stay bit-identical; frames 43–49 are duplicated rows. A policy trained on it learns that commands do nothing.

Joint past its limit

DROID ep. 1, frames 141–172: joint 6 reaches 3.959 rad, 207 mrad past the Panda datasheet limit. It may be an FR3 (limit 4.52 rad); the passport records that ambiguity.

Field doesn't match its docs

DROID's action field is documented as six joint velocities; in all 2,551 frames it equals the commanded Cartesian pose. 8 of 14 episodes carry no language instruction.

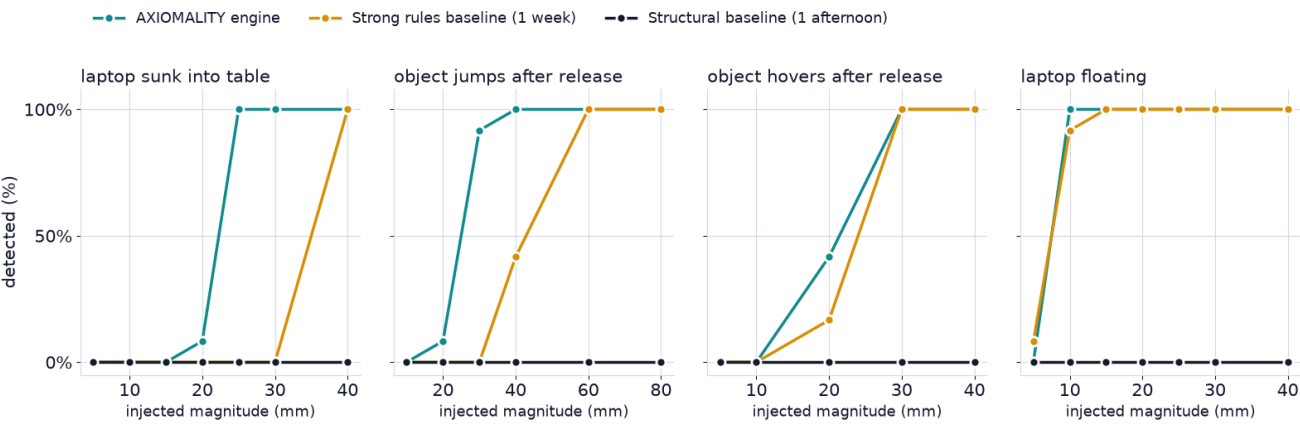
Dataset	Episodes / frames	Accept *	Repair	Reject	Not checked	Passport
DROID (Franka)	14 / 2,551	5	1	8	Timing, semantic	Verified offline
Jaco Play (Kinova)	13 / 879	13	0	0	Timing, semantic	Verified offline
NYU ROT (xArm)	14 / 440	14	0	0	Timing, semantic	Verified offline

Real, public data (OXE public bucket). *Accept covers checkable families only: these datasets record no timestamps or object state. On DROID the structural baseline agrees with the engine on 8 of 14 accept/reject calls; the stuck-arm and joint-limit episodes are ones DROID itself labels as failures.

Checks are copyable; a neutral, signed record is not

Noisy synthetic data, 3 task variants, 20 per error class; no checker reads the injection key; precision 1.00 for all three

Where the engine and the strong baseline differ: smallest detectable error



SYNTHETIC noisy episodes; 12 identical clean base episodes per point; calibrated tolerances as in the benchmark. Interpenetration (not shown): 5-30 mm all caught by both rule-based checkers.

	AXIOMALITY engine	Our strong baseline	Structural (afternoon)
Recall	92%	92%	41%
False rejects (140 clean)	0	0	0
Out-of-order clocks restored exactly	20 / 20	0 (discarded)	0 (discarded)
Missing inputs	Returns UNKNOWN	Rejects	Rejects
Per episode	14 ms	1.4 ms	0.6 ms

What we don't hide

Our own strong baseline, written after the engine, matches its recall. Knowing what to check is not a moat: any team that knows the error list can write the checks.

Where the engine still wins

Finer detection: sinks from 25 mm (baseline 40 mm), post-release jumps from 30 mm (baseline 60 mm). Exact repair of out-of-order clocks. UNKNOWN instead of false rejects. Every verdict cites a versioned, hash-bound rule.

So where is the moat

A neutral issuer, re-checkable verdicts, passports that verify offline, and an error library that grows across customers: things no single buyer or seller builds for itself.

HOW IT WORKS

Versioned, hash-bound, consistency-checked rules: anyone can re-derive a verdict

A VLM judge gives you a plausibility score. We give you a trail anyone can re-check.

Top level · TUpper (ISO/IEC 21838-4:2023)
One of three top-level ontologies with an ISO part, alongside BFO and DOLCE; our founder is a core contributor

Physical world
Objects, parts, shape, measurement, materials, process

Robot manipulation
Affordances, contact events, task phases, joint limits, object permanence

Your episodes
Frames, sensors, lineage; LeRobot / RLDS / MCAP

Rulebook self-check (in the prototype)
30 rules: 24 proven jointly consistent with Z3 on a bounded 14-frame model (one success, one slip); 9 are decided by Z3 per episode on synthetic data; 6 are numeric. Real-data mode runs numeric checks only today.

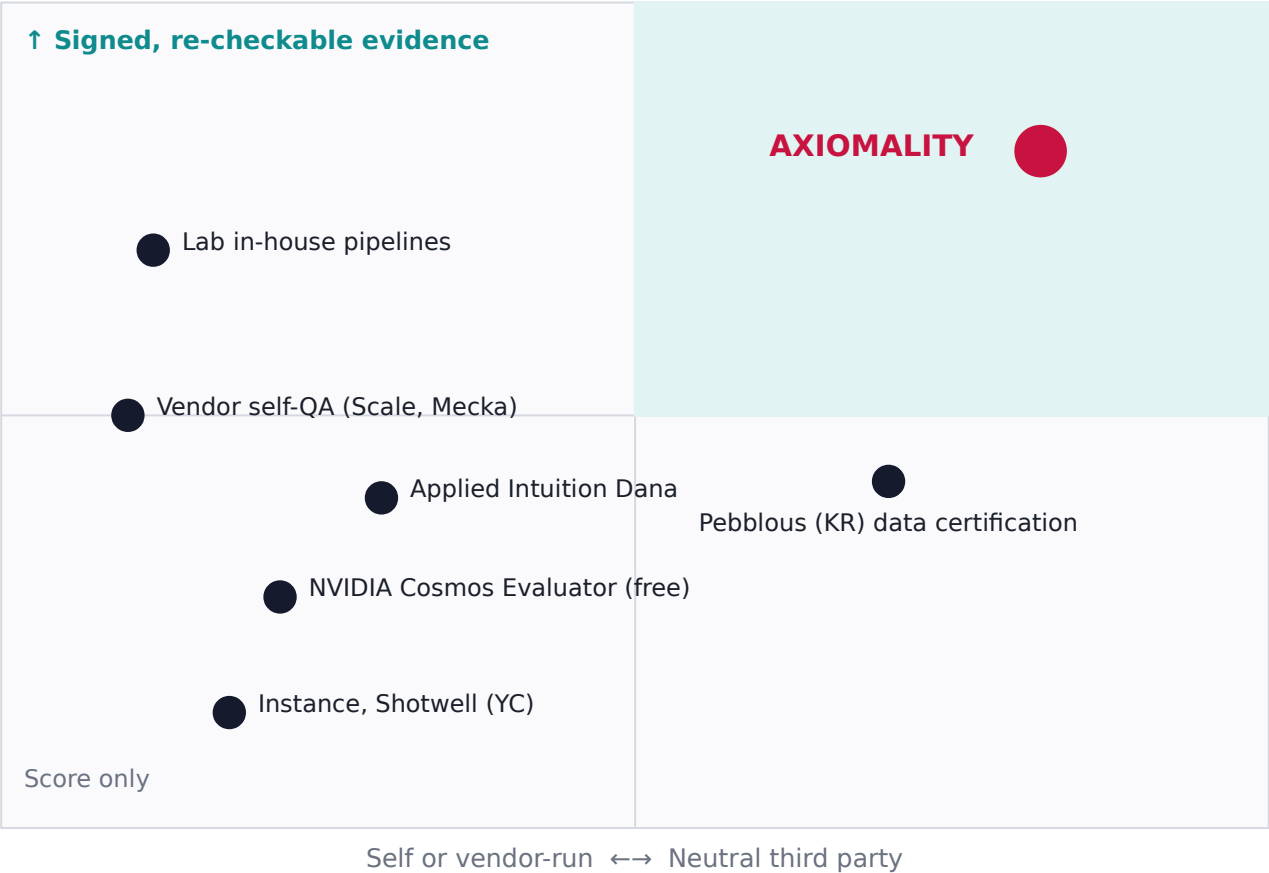
	VLM judge / learned scorer	AXIOMALITY verdict
Output	A score per clip	Accept / repair / reject / unknown per episode
Reason	Opaque	Frame index + rule ID + evidence
Re-checkable by a third party	Re-runnable with the same weights; no rule-level reason	Yes: same rulebook + data hash reproduce the verdict
Missing inputs	Still returns a score	Returns “unknown” and lists what is missing
Improves by	Retraining	Adding a rule, proven consistent before release

The error library we keep (planned, not in the prototype)
Rejects cluster into candidate rules; each is proven consistent with the rest before release, and older batches are re-verified. The moat hypothesis: the rulebook stays consistent and citable at hundreds of rules across task families.

COMPETITION

Everyone scores data; we found no one signing neutral per-episode verdicts

No one signs per-episode physics and logic checks for robot data. The tools where data lives are our rails, not rivals.



vs. NVIDIA

Cosmos Evaluator scores plausibility; it does not sign, does not give rule-level reasons, and lives in one stack. A stack vendor cannot credibly be the neutral referee. We take its score as one input.

vs. in-house QA

In-house QA needs no signature today. It does once one intake gate must hold vendor, in-house and synthetic data to one standard. That gate is where we aim; vendors are the first door.

Rails, not rivals

Foxglove, Rerun and the LeRobot hub are where data lives. The passport ships as their “verified” badge.

Placement is AXIOMALITY’s assessment from public product documentation (2025–26).

Vendors pay per certified dataset; buyers verify for free

Revenue scales with robot-data volume that changes hands, not with consulting hours

Paid pilot
\$25k-50k · 8-12 weeks

One vendor, one task family, their own batch;
blind bake-off against their current scripts;
credited to year one

Per certified hour
Target 5-8% of data price

Vendor pays to ship with a passport; buyer
verifies free with the public key, which pulls the
next vendor in

In-VPC license
Annual

Large vendors and labs run the engine in their
own environment with rulebook updates

Market = robot data changing hands × inspection take rate (model)

Scenario	Hours traded / year	Avg price	Data spend	Take rate*	Inspection revenue pool
Today’s pace	1M	\$100/h	\$100M	5%	\$5M
Industry target	10M	\$100/h	\$1B	6%	\$60M
Synthetic + real, 2030	10M real + 100M synthetic	\$100/h real, \$5/h synthetic	\$1.5B	8%	\$120M

What makes it venture scale

Third-party inspection of traded data alone is a \$5–120M pool. Venture scale needs the intake gate for all training data, in-house and synthetic included; the lab bake-off tests exactly that. Later: the same passport becomes OEM compliance evidence (EU 2028).

Model, not a forecast. Inputs: ~10M-hour industry estimate of data needed; teleop pricing \$50–200/h (DreamVu); Scale 150k+ h (2025). *Take rates are hypotheses the pilots test; synthetic volume is an assumption.

Start with challenger vendors who need to beat Scale on trust

The buyer is the vendor’s head of data ops; the proof is a blind bake-off on their own batch

Target accounts (not customers)	Type	Why they need a passport
Mecka	Teleop data vendor	Scaling fast; needs to prove quality to lab buyers
XDOF	Data pipeline + teleop	Selling to labs at a \$1.2B valuation; trust is the product
Build AI	Egocentric video	Free data needs a quality story to monetize premium tiers
Human Archive	Egocentric data vendor	Seed-stage; differentiation against larger suppliers
Lightwheel	Synthetic / sim data	Synthetic data needs independent acceptance
Labs buying data	Model developer intake	Accept passports at intake; verification is free

Blind bake-off

Same batch through four baselines: the vendor’s scripts, simple rules, VLM + rules, our engine.

Expert ground truth built independently; the customer holds a blind set.

Metrics: critical false-accepts, good-data false-rejects, unknown rate, review hours per 1,000 episodes.

If simple rules match us, we narrow to the error classes they miss.

Then flip the buyer side

Once one lab accepts passports at intake, every vendor selling to that lab has a reason to ship with one.

Accounts are targets chosen from public information; none is a customer or partner today.

What's built, and three things you can check by Dec 31

Everything on this slide can be demonstrated live



Working prototype

- Real data: 41 episodes, 3,870 frames from DROID, Jaco Play and NYU ROT; in DROID it found a stuck arm, an over-limit joint and a mis-documented action field
- Noisy synthetic benchmark: 92% recall, 0 false rejects; ties our own strong baseline, which could close the remaining detail gaps in about a day
- A signed passport for every real dataset, verified offline; change any field and it fails
- One command reproduces everything (run_v2.py)



Foundations

- Company knowledge kernel: formal axioms built on the ISO/IEC 21838-4 top-level ontology
- 100,000 measured 3D asset packages from a prior spatial-computing company, proof we can capture and annotate real geometry
- 13 granted patents; consumer spatial app shipped on Apple Vision Pro



By Dec 31, 2026

- One paid pilot or signed LOI with a data vendor
- One lab's written intake requirement to accept passports
- Rerun on DROID / LeRobot data against a strong hand-written baseline and Cosmos Evaluator; a third party holds the key. If our margin is under 10 points, we narrow to the error classes rules miss

Prototype code and results are in the data room and can be run in one command.

TEAM

People who write machine-checked rulebooks, and have shipped at scale

Formal methods, applied AI, robot perception and 3D capture

Founder & CEO

- Core contributor to ISO/IEC 21838-4, the TUpper top-level ontology; member of the Industrial Ontologies Foundry; University of Toronto AI doctorate and researcher
- Consistency proofs and model construction for first-order axiom systems; 21 papers, invited AAAI talk, one English-language monograph
- Founding team of a cross-border DTC company backed by Accel and others: >\$50M revenue in year one, 100+ people
- Founded a spatial-computing company with one of the first AR apps on Apple Vision Pro; 13 granted patents

CTO & Chief Scientist

McMaster University AI doctorate; former AI scientist at a top-tier tech company and startup chief scientist

Perception & Robotics Lead

Harbin Institute of Technology doctorate in mechatronics; former director of the perception center at XPeng

3D Capture & Simulation Lead

10+ years as art director in games and stage production; led a city-scale VR reconstruction

Hiring now

A robot-learning lead who has trained VLA policies at scale, and a U.S. founding GTM lead for data vendors.

Every robot dataset that changes hands ships with a passport.

1

Vendors ship with passports

2

Labs accept them at intake

3

Synthetic suppliers certify every generated episode

4

OEMs reuse them as compliance evidence



Two intros

One robot-data vendor and one lab procurement lead, for a blind bake-off on their own batch



12 minutes

Corrupt any of our episodes yourself; watch the verdict, then the passport, change



December 31

Check the three commitments on slide 11, one by one